Graph-Clustered Block-QAOA for Deepfake Feature Selection

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Abstract

This study presents an efficient quantum-classical framework for high-dimensional feature selection in deepfake detection. We construct an M-FIG (Multi-Feature Interaction Graph) based QAOA approach, guided by semantic clustering, to optimize the quantum search space and preserve non-linear combinatorial forgery cues. By overcoming the information loss inherent in conventional sequential partitioning, the proposed methodology achieves an 89.56% detection accuracy while utilizing only 25% of the original features.

Introduction & motivation

1. **Increasing Deepfake Realism**
Recent generative AI has made deepfake images highly realistic and harder to detect.

2. **Need for Subtle Feature Analysis**
Detection requires subtle cues such as unnatural skin texture, regional inconsistency, frequency distortion, and generation artifacts.

3. **Limitation of Existing Detectors**
CNN- and ViT-based detectors rely on high-dimensional features, which increase computation, storage cost, and redundancy.

4. **Need for Efficient Feature Selection**
Selecting informative feature subsets can maintain performance while improving efficiency and interpretability.

5. **Optimization Challenge**
Feature selection is a combinatorial optimization problem with an exponentially growing search space.

6. **QAOA with NISQ Limitation**
QAOA is promising for optimization, but direct 512-dimensional optimization is difficult under NISQ constraints.

Our Motivation

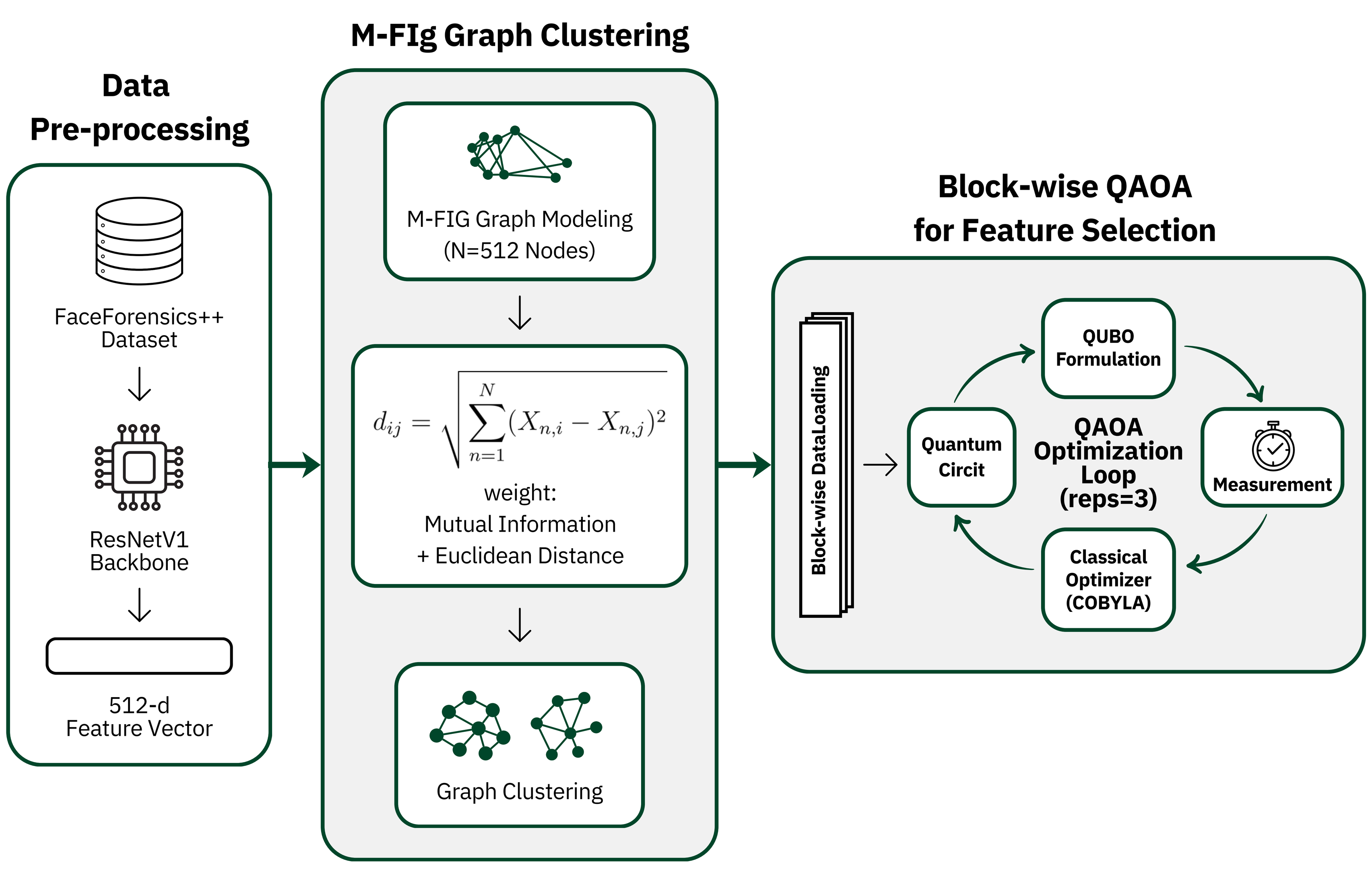
We propose an M-FIG based graph clustering Block-QAOA framework for structure-aware and efficient feature selection.

Research Gap & Our Approach

[Limitations of Previous Paper]	[Proposed Approach]
Pre-filtering [1] → ignores feature interaction	**M-FIG Construction** → models feature relationships as a graph
Linear reduction [2] → weakens nonlinear structure	**Graph Clustering** → groups strongly interacting features
Sequential partitioning [3] → splits related features	**Structure-aware Block-QAOA** → preserves feature relations within blocks

[1] M. Hasan et al., "Quantum-Enhanced Federated Learning via QAOA Feature and Gate Selection with Skew-Symmetric Lipschitz Dynamics," in Proc. Int. Conf. on Computing, Networking and Communications (ICNC), 2026.
[2] Vikas et al., "Quantum-Accelerated Feature Selection for Edge-Enabled Perception in Connected and Autonomous Vehicles," IEEE Access, vol. 13, 2025.
[3] F. Khan et al., "A Quantum-Hybrid Framework for Enhanced Deepfake Detection: Integrating QAOA-Based Feature Selection With Quantum-Inspired Attention Mechanisms," IEEE Access, vol. 14, pp. 17853–17873, 2026.

System pipeline



Core Methodology

1. Clustered Block Construction via Multi-Feature Interaction Graph

Step 1) 512D Feature Graph Modeling
Represent feature interactions as weighted edges

Step 2) Graph Clustering
Group highly correlated features into distinct blocks

Step 3) Block-wise QAOA Selection
Extract the top 256D / 128D informative features

- Preserves semantic feature interactions
- Reduces overall optimization scale

2. Hybrid Quantum-Classical Optimization for Feature Selection

Step 1) QUBO (Quadratic Unconstrained Binary Optimization) Formulation
The optimal feature subset within block B_k is defined by minimizing the objective function:

$$C_k(z) = -\sum_{i \in B_k} w_i z_i + \lambda \sum_{i,j \in B_k, i<j} e_{ij} z_i z_j + \mu \left(\sum_{i \in B_k} z_i - n_k \right)^2$$

Step 2) Hamiltonian Mapping via Ising Symmetry
Transforming the QUBO model into a Cost Hamiltonian using $z_i \rightarrow \frac{1-\sigma_i^z}{2}$

$$H_C^{(k)} = \sum h_i \sigma_i^z + \sum J_{ij} \sigma_i^z \sigma_j^z$$

Step 3) Variational Quantum Ansatz (QAOA)
Applying alternating Cost (H_C) and Mixer (H_M) layers with depth $p = 3$:

$$|\psi(\gamma, \beta)\rangle = \left(\prod_{m=1}^{3} e^{-i\beta_m H_M} e^{-i\gamma_m H_C^{(k)}} \right) |+\rangle^{\otimes |B_k|}$$

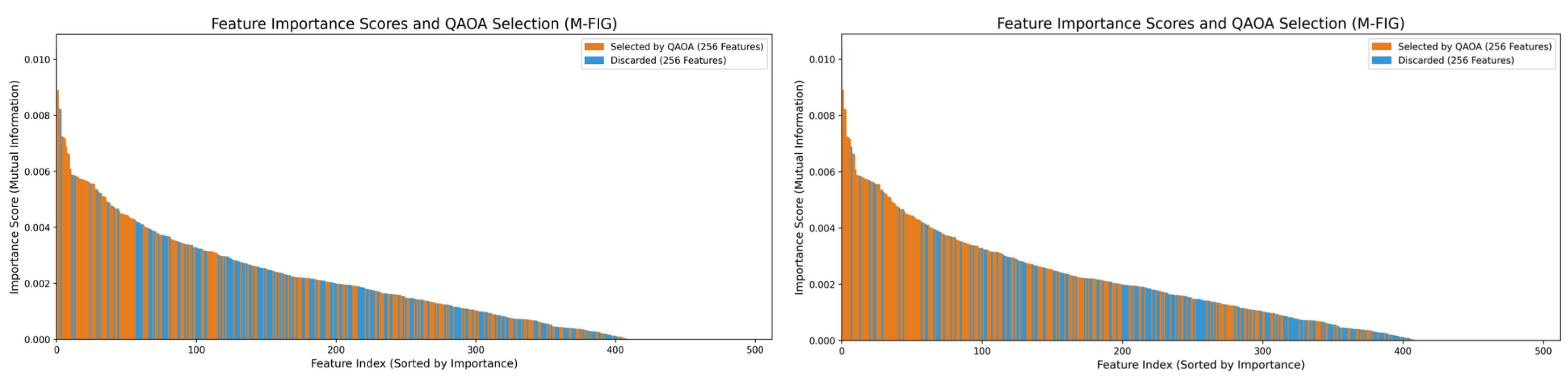
Step 4) Classical Feedback & Feature Aggregation
Optimal parameters derived via a COBYLA-based classical optimization loop.

Key Results

[Experimental Setup]
Dataset: FaceForensics++ (4 forgery techniques, 180,000 images)
Feature Extraction: ResNetV1 (512 dimensions)
Quantum Simulation: IBM Qiskit (StatevectorSampler)

Method	Num of Features	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
ViT (Attn SOTA)	512	85.8	85.6	99.21	91.9
ResNet+CBAM	512	89.56	91.45	96.13	93.73
SRM+CNN (Freq)	512	90.05	91.22	97.09	94.07
Sequential QAOA	256	88.93	91.64	95.04	93.31
Proposed 1	**256**	**89.27**	**93.46**	**93.31**	**93.39**
Proposed 2	**128**	**89.56**	**92.36**	**93.66**	**93.66**

Even if a feature is identified as highly important, it is boldly removed if it shares a redundant activation pattern with other selected features.



[Key Achievements]
Compression Efficiency: Achieved 89.56% accuracy without performance degradation using only 25% of the original features.
Computational Acceleration: Reduced optimization convergence time per block by ~82% through structural reduction of the problem space.
Enhanced Reliability: Improved precision (92.36%) and significantly suppressed False Positive rates by eliminating redundant information.

Conclusion & Future Work

By preserving the 'combinatorial context of data' prior to quantum optimization, the M-FIG-based structural space partitioning effectively overcomes the physical constraints of NISQ environments and drastically improves the efficiency and generalization performance of deepfake detection models.
Conduct cross-validation and evaluate generalization performance on unseen forged images generated by state-of-the-art generative models (e.g., Diffusion models).